making one DB call (even if mediated through a fast cache) is likely going to be too slow, we probably want to have another daemon running on our server's machine on a thread in the application server that gets the diff from the database, and only for the values changed from last update, goes through the list of shops (kept in some shared memory area) and updates those values.

11.3.3 Lessons Learned

We have delved into our e-commerce application, iterating from a coarse-grained design for a centralized application up to the smallest details of a distributed system.

While this discussion can't and doesn't mean to be exhaustive, I hope it was useful to provide you with an idea of how you can structure the design process that leads from an algorithm to a full, production-ready application that leverages it; hopefully it also provided useful pointers to the possible issues you could face in the development of a web application, and what you could look for if you'd like to keep learning on this area.

Now it's time to move on and present you a few more problems, in completely different contexts, that can be solved using nearest neighbor search.

11.4 Other Applications

Maps aren't the only field of application for nearest neighbor search; they were not even the application for which k-d